

Module 4

Clodu Storage in GCP



Goals: Understand about Cloud Storage

- Create bucket
- Use case of each storage class
- Object access control
- Protect object data, object versioning

- **Google Cloud Storage** is a **RESTful** online file storage web service for storing and accessing data on **Google Cloud Platform** infrastructure.
 - Cloud Storage is unified object storage service.
 - Cloud Storage is a persistent storage, it is durable, replicated and also made globally available via HTTP URL.
 - Cloud Storage is **auto scalable** service.
 - Each item in cloud storage have unique URL.
 - Store all file types - text, binary, backup & archives: Media files and archives, Application packages and logs
- Backups of your databases or storage devices
- Staging data during on-premise to cloud database migration

► Key Terms:

- **Buckets**: Basic containers that hold your data. Everything that you store in Google Cloud Storage must be contained in a bucket. You can use buckets to organize your data and control access to your data, but unlike directories and folders, you cannot nest buckets.
- **Bucket names**: Should be unique as the name of the buckets stored in single Cloud Storage namespace. Also, bucket names can be used with a CNAME redirect, which means they need to conform to DNS naming conventions.
- **Bucket labels**: Bucket labels are key:value metadata pairs that allow you to group your buckets along with other Google Cloud.

► Key Terms:

- **Objects**: Objects are the individual pieces of data that you store in Google Cloud Storage.
- Objects have two components: object data and object metadata.
- The object data component is usually a file that you want to store in Google Cloud Storage.
- The object metadata component is a collection of name-value pairs that describe various object qualities.
- There is no limit on the number of objects that you can create in a bucket

Object versioning:

- Prevents accidental deletion & provides history
- Enabled at bucket level
- Can be turned on/off at any time
- Live version is the latest version
- If you delete live object, it becomes noncurrent object version
- If you delete noncurrent object version, it is deleted
- Older versions are uniquely identified by (object key + a generation number)
- Reduce costs by deleting older (noncurrent) versions!

Module 4: Cloud Storage

	Multi Regional	Regional	Nearline	Coldline
Intended for Data	Most Frequently Accessed	Accessed Frequently within a Region	Accessed less than once a month	Accessed less than once a Year
Availability SLA	99.95%	99.90%	99.00%	99.00%
Access APIs	Consistent			
Access Time	Milisecond Access			
Use Cases	Content Storage & Delivery	In-region Analytics, Transcoding	Long-tail contents, Backups	Archiving, Disaster Recovery

Module 4: Cloud Storage

Storage Class	Name	Minimum Storage duration	Typical Monthly availability	Use case
Standard	STANDARD	None	> 99.99% in multi region and dual region, 99.99% in regions	Frequently used data/Short period of time
Nearline storage	NEARLINE	30 days	99.95% in multi region and dual region, 99.9% in regions	Read or modify once a month on average
Coldline storage	COLDLINE	90 days	99.95% in multi region and dual region, 99.9% in regions	Read or modify at most once a quarter
Archive storage	ARCHIVE	365 days	99.95% in multi region and dual region, 99.9% in regions	Less than once a year

Quizzes

- How do you speed up large uploads (example: 100 GB) to Cloud Storage?
- You want to permanently store application logs for regulatory reasons. You don't expect to access them at all.
- Log files stored in Cloud storage. You expect to access them once in quarter.
- How do you change storage class of an existing bucket in Cloud Storage?

Answers

1. Use Parallel composite uploads (File is broken into small chunks and uploaded)
2. Cloud storage – Archive
3. Cold Line
4. Step 1: Change Default Storage Class of the bucket.
Step 2: Update the Storage Class of the objects in the bucket.

Thank you

